



**Ghulam Ishaq Khan Institute of Engineering
Sciences
and Technology (GIKI)**

**Project Scope Document
for**

Project : Device-Free Fall Detection Using WiFi CSI

Version 1.0

by

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Abstract

Elderly falls in residential and care facilities represent a safety issue of paramount importance that needs to be addressed urgently. The time taken before the person is helped by the other increases the threat of serious health complications in case of a fall. The conventional monitoring systems do not cope with this well: wearable technology is poorly received because of discomfort or cognitive issues, and camera surveillance is not accepted in most cases as an invasion of privacy, especially in private spaces such as bedrooms and bathrooms where falls are common.

This project focuses on eliminating these core constraints by switching to a data-driven, device-free sensing scheme, based upon the previously manual or visual monitoring. The proposed system takes advantage of the WiFi Channel State Information (CSI) to identify physical disturbances in the environment without the need of wearable devices or capturing images. The architecture records high-resolution radio-frequency data extraction, where a normal WiFi router is a transmitter, and an Intel development board Network Interface Card is a receiver. The raw signal data is optimally processed using a deep learning pipeline, [3], [4], by extracting dynamic spatial and temporal features, i.e., CSI Amplitude and Phase Difference. The project checks advanced architectures to precisely differentiate the distinct impact signature of a fall and usual activities of daily living (ADL). The main goal is to send the automated, real-time notifications to caregivers when the incident is detected, which will significantly decrease the emergency response time. This privacy-first, scale-able solution will protect vulnerable residents and preserve their overall dignity and autonomy.

1. Introduction

The welfare of the vulnerable group, the elderly and the unsupervised children are a major concern in the residential healthcare environment between families and healthcare facilities. Monitoring is a key to the speed of medical response in case of accidents, but the conventional home safety practices are becoming limited in their application due to the privacy issues, prohibitive cost, and non-conformity to use by the users. To ensure the dignity and autonomy of the elderly, they should offer automated and minimal-invasive solutions that will save on the labor force and have maximum safety.

Traditional methods of residential security are usually based on manual control, wearable panic buttons or in-place camera surveillance. These solutions impose major constraints: manual inspections cause dangerous time lapses during which a fallen person can go unattended, wearables are often not remembered, and camera-based systems invade the privacy of intimate locations such as bedrooms and bathrooms. To mitigate these severe shortcomings, our suggested system takes advantage of WiFi Channel State Information (CSI) sensing technology [3] to measure movement dynamics and sense falls, yet not capture them visually. With the help of ambient radio-frequency infrastructure, this project provides a solution that is practical, privacy-conscious, and cost-effective at the same time to manage the home safety efficiently [1], [2]..

2. Problem Statement

Within the residential environment of the contemporary care facilities and households, the elderly are highly challenged with the danger of unnoticed fall. In these settings, the staff to resident ratio can be such that one-on-one observation cannot be sustained, and thus an immediate response to an incident is of the utmost importance. The current safety measures mainly consist of manual observations through regular staff inspections or simple alert systems like pull-cords and fixed emergency buttons. These techniques are not very effective since they depend on the fallen person to be awake and physically able to seek aid, which can very often cause a significant delay in communication and additional physical effort on the body. Caregivers usually adhere to the regular observation schedules that do not mention the unstable time of fall that occur at different days and at different times depending on the activities and mobility of the resident. This ineffectiveness not only affects the personal recovery of the person, but also puts a strain on the operational resources and time management of the care facility. Moreover, manual check-ins require a lot of staff labor, which requires an employee to juggle between various tasks of caring. Although there are a few automated options, [4] they are often either too costly to be used by many or too complicated such that they require intrusion cameras that breach privacy or wearable gadgets that the residents often forget to use, especially those with cognitive impairments..

3. Problem Solution/Objectives of the Proposed System

The necessity to have a system that optimizes fall detection and safety monitoring in residential settings with full privacy, resource optimization and better health results is evident. Our solution is a device-free one, where the caregiver gets to automate safety monitoring on the basis of real radio-frequency environmental measurements, instead of depending on manual efforts, guesses, and invasive visual surveillance.

The system will be based on the use of inexpensive and wifi sensing chipset to track the WiFi Channel State Information (CSI) in a room. The system uses the differences in subcarrier amplitude and phase to detect the particular signal signature of a human fall [3]. This crude data is then run through a smart deep learning pipeline [4] with architectures that make real time decisions about emergency events. The platform will provide the timely delivery of help by implementing automated delivery of alerts via mobile or web interfaces, which means that help can be delivered when a situation happens. Also, the hardware will be of a strong and low-maintenance nature, so that the system is not affected easily and it can be used in everyday households.

3.1 Objectives

- **BO-1: Wearable-Free Data Capture:** Continuously collect high-granularity WiFi CSI data using low-cost edge sensors without using any wearables.
- **BO-2: Deep Learning Classification:** Train an effective model to correctly distinguish between real falls and everyday activities by signal pattern.
- **BO-3: Real-Time Edge Processing:** Implement the detection algorithm on the edge hardware locally to reduce latency and issue automatic alerts as soon as possible.
- **BO-4: Privacy Preservation:** Monitor movements based only on ambient WiFi noise, and no cameras or wearables are necessary.
- **BO-5: Scalability:** Develop a prototype that is affordable so that standard care facilities can use it.
- **BO-6: Environmental Resilience:** Be able to maintain a consistent detection accuracy even under normal indoor conditions such as ADL motion or size variations.
- **BO-7: Caregiver Efficiency:** Implement an efficient alert system that will cut down on the amount of manual monitoring and allow quick emergency response.

4. Related System Analysis/Literature Review

Table 1 : Related System Analysis with proposed project solution

Application Name	Features	Weakness	Relevance with the Proposed System
Wearable-based Detectors	• Direct acceleration measurement • Low-cost hardware	• Requires active user compliance • Frequent battery maintenance	• Entirely wearable-free approach • Passive monitoring via WiFi signal analysis
Vision-based Systems [5]	• High-resolution visual confirmation • Strong spatial accuracy	• Severe privacy intrusion • Fails in low-light or blocked views	• Maintains total privacy • Operates via signal disruptions rather than images

CSI-based WiFi Sensing[3], [4]	• Standard router implementation • Uses simple signal metrics	• Coarse data granularity • Sensitive to environmental noise	• Utilizes high-granularity CSI data • Uses deep learning on changes in frequency
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5. Vision Statement

To the elderly population who needs a trusted safety monitoring system and yet does not interfere with their personal privacy, our system provides an automated, wearable-free detection system that will provide a balance between human supervision and data-driven security. Such an intelligent safety solution is the only way to ensure that the residents are notified immediately after a fall and at the same time, their dignity is preserved unlike the traditional ways of monitoring falls using intrusive cameras or hardware that is easily lost. The system would create a scalable and economical safety infrastructure of the current care facilities by dramatically lowering response times and reducing the physical burden linked with slow responses.

6. Scope

The proposed Fall Detection System is uniquely created to be used in domestic home settings, emphasizing the safety requirements of at-risk individuals and not infringing on privacy. The system boundaries include the deployment of low cost hardware, the creation of signal processing pipeline, and the incorporation of a deep learning classification model.

The essence of the project is based on the radio-frequency data capture and analysis. The system uses an intel development board Network Interface Card which serves as a receiver and a standard router as transmitter.. These will be strategically located in a room to create a sense connection and they will be independent of the main internet router of the house to provide high frequency and continuous flow of data.

The scope contains key features such as:

- **CSI Data Acquisition:** Raw Channel State Information (64 subcarriers) at a constant sampling rate (e.g., 100Hz) with custom embedded C++ firmware. [1]
- **Signal Preprocessing:** Noise reduction goes through to eliminate unchanging environmental and hardware noise..
- **Feature Extraction:** Extraction of CSI Amplitude and Phase Differences to effectively measure the spatial and temporal properties of human movement.
- **Deep Learning Classification:** Training time-series AI models to correctly classify a "Fall" and normal "Activities of Daily Living (ADL)." [4].
- **Automated Alerts:** Creating a dashboard or mobile notification system that automatically notifies the caregivers whenever a fall is detected.
- **Edge Deployment:** Making the software architecture such that the trained Deep learning model can be executed locally on local systems at zero-latency to monitor.

7. Project Stakeholders and Roles

Table 2: Project Stakeholders for [PROJECT]

Project Sponsor	Ghulam Ishaq Khan Institute (GIKI)
Stakeholder	• Sarosh Ishaq • Muqem • Mehreen • Dr Muhammad Salman • Senior Design Project Evaluation Panel

8. References

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